Thadomal Shahani Engineering College, Bandra
Department of Artificial Intelligence & Data Science

Name:Rashi Bhatia				**Subject: ASD&D**
TE - 1	**Batch: T11**	**Sem: 5**	**Roll No:12**	**Date:28/7/2026**

Experiment 3

AIM : To deploy multi-service applications using Docker Compose.

LO MAPPED : 03

LO: Build and manage automated CI/CD pipelines for application development,

testing, packaging and deployment.

THEORY:

What is yaml?

YAML is a human-readable data serialization language primarily used for writing software configuration files.It stands for YAML Ain't Markup Language and acts as a user-friendly alternative to XML and JSON.Structure is defined strictly through line breaks and indentation, entirely eliminating the need for brackets, braces, or tags.It supports native data types like lists, maps, strings, and booleans, while also allowing inline comments using the # symbol.Today, it is the universal industry standard for managing Docker, Kubernetes, and CI/CD pipeline automation scripts.

Managing applications with multiple microservices (e.g., a frontend web server, a backend API, and a database) using individual docker run commands becomes highly complex, error-prone, and difficult to maintain. **Docker Compose** is a tool specifically designed for defining and running multi-container Docker applications. Instead of configuring each container manually, you define your entire application stack in a single configuration file.

2. Core Concepts

- **YAML Configuration File (docker-compose.yml)**: Docker Compose uses a human-readable YAML format to define the **services**, **networks**, and **volumes** required for the application stack.
- **Services**: A service represents a containerized component of your application (e.g., web, db). It references a specific Docker image or a local build context.

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- **Container Orchestration**: With a single command (docker compose up), Docker Compose handles the creation, initialization, and configuration of all interrelated containers simultaneously.
- **Service Isolation and Networking**: By default, Docker Compose sets up a single default **bridge network** for your app. Each container for a service joins this network, allowing them to communicate with each other using the service name as the hostname (via automatic DNS resolution).
- **Data Persistence**: **Volumes** are declared within the configuration file to map host directories to container paths, ensuring database entries and asset uploads persist even if containers are destroyed and recreated.
- **Dependency Management**: Using the depends_on parameter, developers can specify deployment hierarchies, ensuring that foundational services (like databases) initialize before dependent application layers start up.

CODE :

```
version: '3.8'
services:
  web:
    image: nginx
    container_name: webserver
    ports:
      - "8080:80"
  database:
    image: mysql:8.0
    container_name: mysqlserver
    environment:
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_DATABASE: studentdb
      MYSQL_USER: student
      MYSQL_PASSWORD: student123
    ports:
      - "3307:3306"
    volumes:
      - mysql-data:/var/lib/mysql
volumes:
  Mysql-data:
```

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OUTPUT :

docker compose up -d:

Starts your containers in the background so you can keep using the terminal.

```
AI&DS203@LAB05-020 MINGW64 ~/Documents/compose-demo (main)
$ docker compose up -d
[+] Running 2/2
 ✓ Container webserver    Running
 ✓ Container mysqlserver  Started
```

docker compose ps:

Lists all running containers for the application along with their current status.

```
AI&DS203@LAB05-020 MINGW64 ~/Documents/compose-demo (main)
$ docker compose ps
NAME          IMAGE       COMMAND                  SERVICE    CREATED              STATUS              PORTS
mysqlserver   mysql:8.0   "docker-entrypoint.s…"   database   22 seconds ago       Up 21 seconds       33060/tcp, 0.
0.0.0:3307->3306/tcp
webserver     nginx       "/docker-entrypoint.…"   web        About a minute ago   Up About a minute   0.0.0.0:8080-
>80/tcp
```

docker compose down:

Stops and completely removes the application containers, networks, and internal volumes.

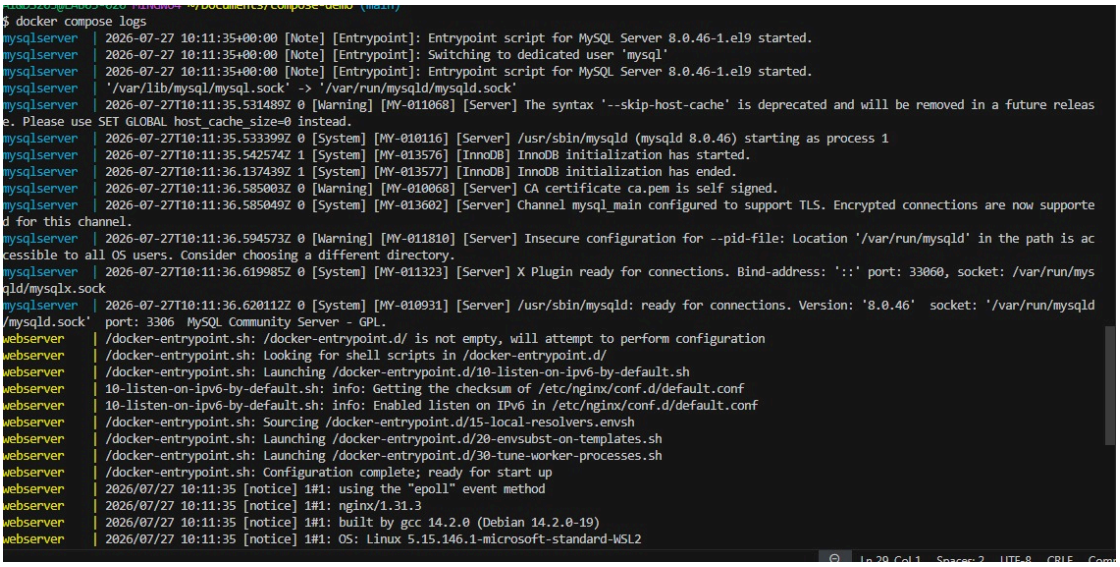
```
AI&DS203@LAB05-020 MINGW64 ~/Documents/compose-demo (main)
$ docker compose down
[+] Running 3/3
 ✓ Container webserver           Removed
 ✓ Container mysqlserver         Removed
 ✓ Network compose-demo_default  Removed
```

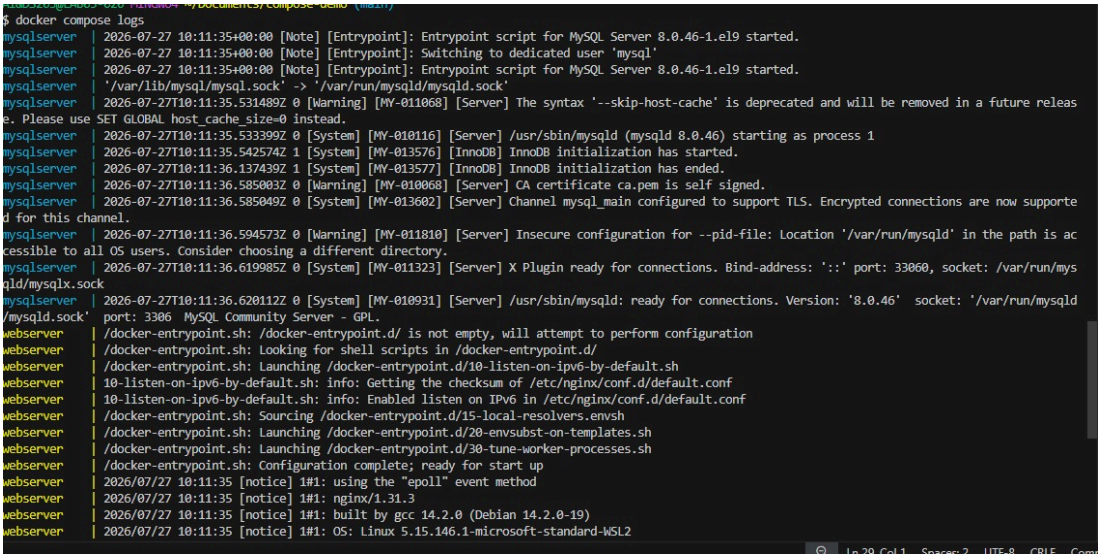
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TE - 1	Batch: T11	Sem: 5	Roll No:12	Date:28/7/2026

docker compose logs:

Displays the aggregated log output from all the services in your application.

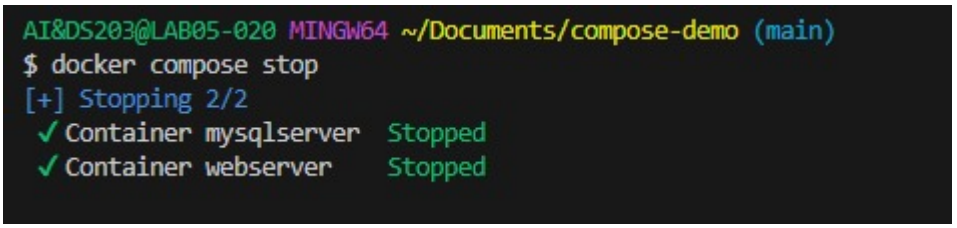




Name:Rashi Bhatia				Subject: ASD&D
TE - 1	Batch: T11	Sem: 5	Roll No:12	Date:28/7/2026

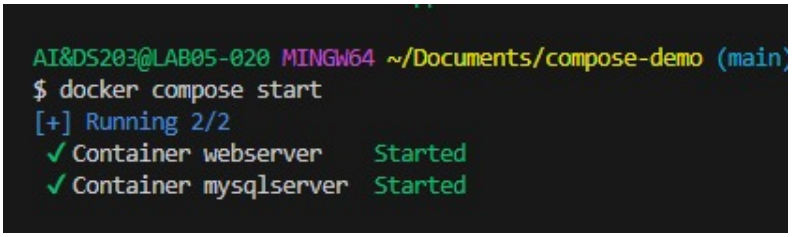
docker compose stop:

Pauses the running containers without removing them or deleting any network data.



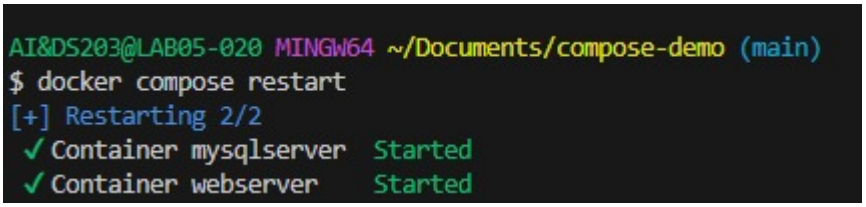
docker compose start:

Restarts any existing containers that were previously paused by the stop command.



docker compose restart:

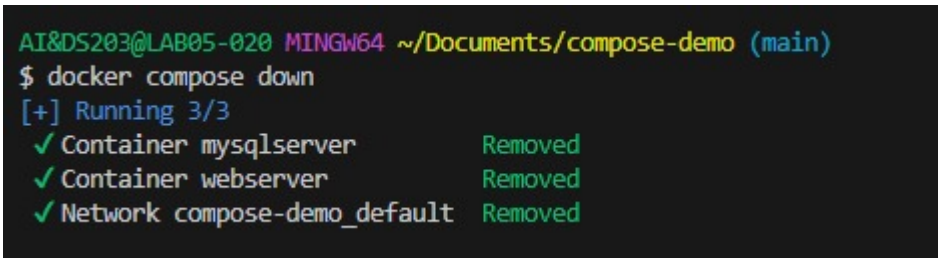
Stops and immediately starts all the application containers to refresh their state.



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TE - 1	Batch: T11	Sem: 5	Roll No:12	Date:28/7/2026

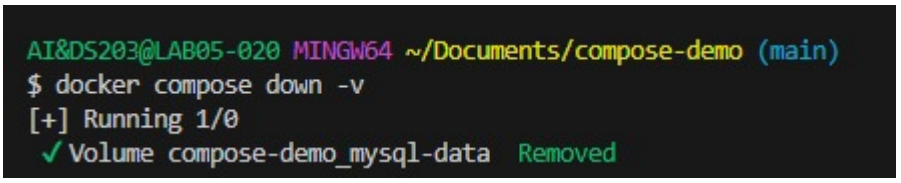
docker compose down:

Stops and completely removes the application containers, networks, and internal volumes.



docker compose down -v:

Stops containers and additionally deletes all attached persistent data volumes permanently.



docker compose images:

Shows a list of the container images used by your current services.

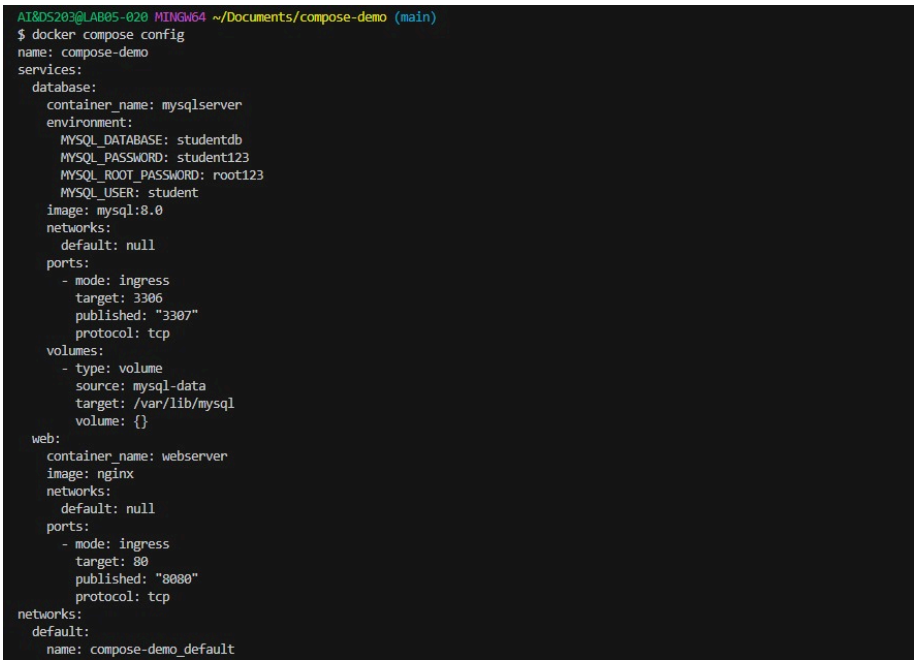


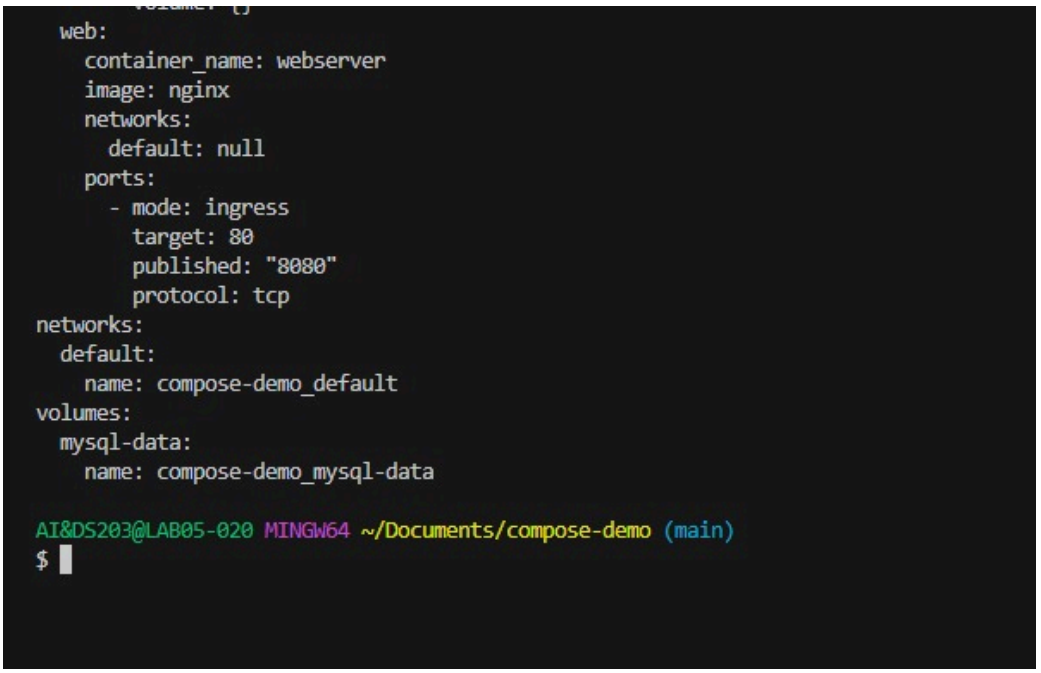
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docker compose config:

Validates the YAML syntax and displays the final, compiled configuration file. [1, 2, 3, 4, 5]





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Conclusion

- **Successful Deployment**: The experiment successfully demonstrated the deployment of a multi-service application stack using **Docker Compose** rather than executing fragmented, isolated container runs.
- **Simplified Workflows**: Consolidating multi-container infrastructure into a single docker-compose.yml file significantly reduces human error, unifies environment variables, and enables single-command orchestration via docker compose up and docker compose down.
- **Service Interaction**: The application demonstrated seamless data persistence and inner-container communication, confirming that Docker Compose effectively automates network bridging and internal DNS mapping using designated service keys.
- **Final Takeaway**: Ultimately, Docker Compose serves as a critical bridge between local application development and production orchestration, standardizing environments and ensuring reproducible deployments across varying computing infrastructures.